Project Report

The current mirror qubit

Guide: Prof. Sudhir Ranjan Jain

A R Bathri Narayanan
Roll no: P0211501
UM DAE Centre for Excellence in Basic Sciences

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Certificate

This is to certify that CEBS student **A R Bathri Narayanan** has undertaken project work from **06/01/2025 to 01/05/2025** under the guidance of **Prof. Sudhir Ranjan Jain**
This submitted project report titled **The current mirror qubit** is towards the academic requirements of the M.Sc. program's 8th Semester Project Course at UM-DAE CEBS.

A. R. Bathri Narayanan
01/01/2025

A R BATHRI NARAYANAN, P0211501
Name & Signature of Student with Date

Sudhir Ranjan Jain May 1, 2025
Name & Signature of Advisor (s) with Date

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Abstract

The current mirror qubit is a popular concept due to it's protection from relaxation and dephasing due to it's structure. In this project, we study the properties of the same, examples include the Lagrangian, Hamiltonian and the equation of motion for the system.

We also form a method to obtain the wave equation for the system, by transforming into the exciton and agiton subspace, and finding suitable solutions for the signal to propagate through the qubit.

Contents

Chapter 1

Current mirror qubit: An introduction

The current mirror qubit is an array of Josephson junctions, 2N of them, but then there is a twist in the end, like a Möbius strip. The upper and lower stripes are connected by capacitors, thus forming N units, as seen in the figure below. the Josephson junctions have the capacitance as C_J and Josephson energy $E_J = \hbar I_c/2e$. We also have ground capacitance C_g attached to each node, reduced flux variables $\phi_i = 2\pi\Phi_i/\Phi_0$ where $\Phi_0 = h/(2e)$ is the superconducting flux quantum and Φ_i is the generalised flux variable.

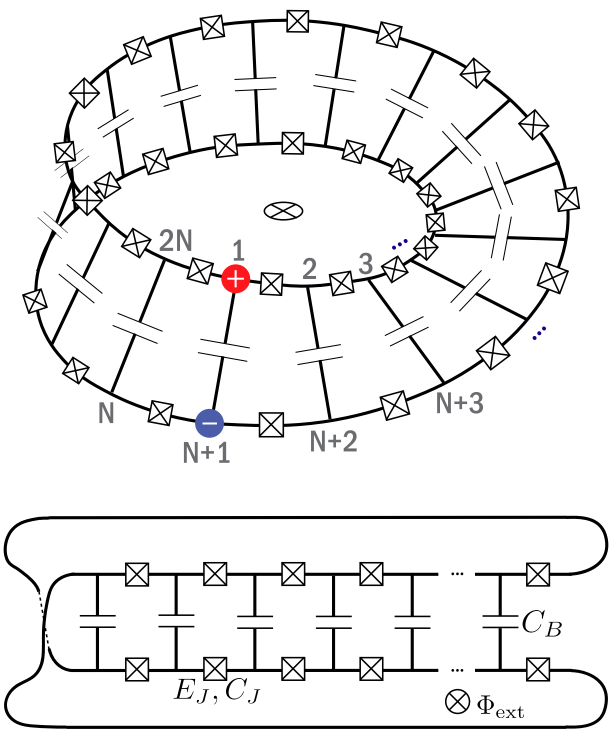


Figure 1: The Current Mirror Qubit as defined in [1]

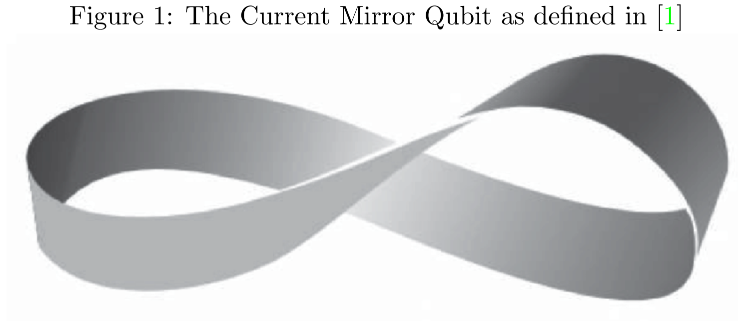


Figure 2: The Möbius strip (Picture taken from [2])

1.1 Equation of Motion of the system

The Lagrangian of the system is given by[1]

$$\mathcal{L} = \frac{1}{2}\left(\frac{\phi_0}{2\pi}\right)^2 \sum_{i,j=1}^{2N} \dot{\phi}_i C_{ij} \dot{\phi}_j + \hbar \sum_{j=1}^{2N} \dot{\phi}_j n_{gj} + \sum_{j=1}^{2N} E_J \cos\left(\phi_{j+1} - \phi_j - \frac{\phi_{ext}}{2N}\right) \tag{1.1}$$

where

$$\mathsf{C}_{ij} = \begin{cases} C_g + 2C_J + C_B, & i = j \\ -C_J, & i = j \pm 1 \\ -C_B, & i = j \pm N \\ 0, & \text{otherwise.} \end{cases} \tag{1.2}$$

Trying to obtain the Euler Lagrange equation of motion for the system,

$$\frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}_j}\right) = \frac{1}{2}\left(\frac{\phi_0}{2\pi}\right)^2 \sum_{i=1}^{2N} \ddot{\phi}_i C_{ij} + \hbar \dot{n}_{gj} \;\; ; \;\; \frac{\partial \mathcal{L}}{\partial \phi_j} = E_J \sin\left(\phi_{j+1} - \phi_j + \frac{\phi_{ext}}{2N}\right) - E_J \sin\left(\phi_j - \phi_{j-1} + \frac{\phi_{ext}}{2N}\right)$$

and equating the two, we obtain the equation of motion for the system

$$\frac{1}{2}\left(\frac{\phi_0}{2\pi}\right)^2 \sum_{i=1}^{2N} \ddot{\phi}_i C_{ij} + \hbar \dot{n}_{gj} = E_J \sin\left(\phi_{j+1} - \phi_j + \frac{\phi_{ext}}{2N}\right) - E_J \sin\left(\phi_j - \phi_{j-1} + \frac{\phi_{ext}}{2N}\right) \tag{1.3}$$

Expanding the matrix elements, we finally get the equation of motion as

$$\frac{1}{2}\left(\frac{\phi_0}{2\pi}\right)^2 \left[(C_g + 2C_J + C_B)\ddot{\phi}_j - C_J \ddot{\phi}_{j\pm1} - C_B \ddot{\phi}_{j\pm N}\right] + \hbar \dot{n}_{gj} = E_J \sin\left(\phi_{j+1} - \phi_j + \frac{\phi_{ext}}{2N}\right) - E_J \sin\left(\phi_j - \phi_{j-1} + \frac{\phi_{ext}}{2N}\right) \tag{1.4}$$

1.2 Hamiltonian of the system

We take the Lagrangian 1.1 and try to find the conjugate momenta for each flux variable.

$$\frac{\partial \mathcal{L}}{\partial \dot{\phi}_j} = \Pi_j = \frac{1}{2}\left(\frac{\phi_0}{2\pi}\right)^2 \sum_{i=1}^{2N} \dot{\phi}_i C_{ij} + \hbar n_{gj}$$

We take a frame such that the offset charge can be neglected, finding conjugate momentum for each case, taking $\frac{1}{2}\left(\frac{\phi_0}{2\pi}\right)^2 = \mu$

$$\Pi_1 = \mu \sum_{i=1}^{2N} \dot{\phi}_i C_{i1}$$

$$\Pi_2 = \mu \sum_{i=1}^{2N} \dot{\phi}_i C_{i2}$$

.....

$$\Pi_N = \mu \sum_{i=1}^{2N} \dot{\phi}_i C_{iN}$$

.....

$$\Pi_{2N} = \mu \sum_{i=1}^{2N} \dot{\phi}_i C_{i2N}$$

Hence, we have a matrix equation

$$\mu C \dot{\mathbf{\Phi}} = \mathbf{\Pi} \tag{1.5}$$

Where C is the capacitance matrix defined by 1.2, $\dot{\mathbf{\Phi}}$ are the 2N flux variable derivatives, and $\mathbf{\Pi}$ are the associated conjugate momenta.
We can finally arrive at the Hamiltonian by

$$\mathcal{H} = \sum_{j=1}^{2N} \Pi_j \dot{\phi}_j - \mathcal{L} \tag{1.6}$$

Modifying the Lagrangian in a slight way by incorporating the constant multiple into the capacitance matrix, making it $\widetilde{C}$

$$\mathcal{L} = \sum_{i,j=1}^{2N} \dot{\phi}_i \left[\widetilde{C}_{ij}\dot{\phi}_j + \hbar n_{gj}\right] + \sum_{j=1}^{2N} E_J \cos\left(\phi_{j+1} - \phi_j - \frac{\phi_{ext}}{2N}\right) \tag{1.7}$$

we get the Hamiltonian of the form

$$H = \sum_{i,j=1}^{2N} 4E_{c_{i,j}} \left(p_i - n_{gi}\right)\left(p_j - n_{gj}\right) - 2\sum_{j=1}^{2N} E_J \cos\left(\phi_{j+1} - \phi_j - \frac{\phi_{ext}}{2N}\right) \tag{1.8}$$

Where E_c is the inverted capacitance matrix. We shall try identifying E_c in the next section

1.3 Continuum Hamiltonian

Taking the offset charges to 0 and Tayler expanding cosine around 0 or π, we get

$$H = \sum_{i,j=1}^{2N} 4E_{c_{i,j}} \left(p_i p_j\right) - 2\sum_{j=1}^{2N} E_J \left[1 - \left(\phi_{j+1} - \phi_j - \frac{\phi_{ext}}{2N}\right)^2\right] \tag{1.9}$$

Taking N to a very large value and assuming smaller steps, we can write the Hamiltonian as

$$H = \int_{x,y=1}^{2N} 4E_c(x,y)\left(\frac{d\phi(x)}{dt}\frac{d\phi(y)}{dt}\right) dxdy - 2\int_{y=1}^{2N} E_J \left[1 - a^2\left(\frac{\phi(y+1) - \phi(y)}{a}\right)^2\right] dy \tag{1.10}$$

$$H = \int_{x,y=1}^{2N} 4E_c(x,y)\left(\frac{d\phi(x)}{dt}\frac{d\phi(y)}{dt}\right) dxdy - 2\int_{y=1}^{2N} E_J \left[1 - a^2\left(\frac{d\phi(y)}{dx}\right)^2\right] dy \tag{1.11}$$

Ignoring the constant terms, we get the final Hamiltonian as,

$$H = \int_{y=1}^{2N} dy \left[\left(4\int_{x=1}^{2N} dx.E_c(x,y)\left(\frac{d\phi(x)}{dt}\frac{d\phi(y)}{dt}\right) dx\right) - 2E_J a^2\left(\frac{d\phi(y)}{dx}\right)^2\right] \tag{1.12}$$

In the next chapter, we devise a way to obtain the E_c matrix.

1.4 Dispersion relation of the system

We make a minor simplification

$$\frac{1}{2}\left(\frac{\phi_0}{2\pi}\right)^2 \left[(C_g + 2C_J + C_B)\ddot{\phi}_j - C_J\ddot{\phi}_{j\pm1} - C_B\ddot{\phi}_{j\pm N}\right] + \hbar\dot{n}_{gj} = 2E_J \cos\left(\frac{1}{2}(\phi_{j+1} - \phi_{j-1}) + \frac{\phi_{ext}}{2N}\right)\sin\left(\frac{1}{2}(\phi_{j+1} + \phi_{j-1}) - \phi_j\right) \tag{1.13}$$

If we set the external flux to 0, and take the continuous approximation at long wavelength (i.e there is not much difference between the consecutive flux variables), and we assume the offset charge doesn't change with time i.e. $\dot{n}_{gj}$=0. We can shorten this equation to

$$\frac{1}{2}\left(\frac{\phi_0}{2\pi}\right)^2 \left[(C_g + 2C_J + C_B)\ddot{\phi}_j - C_J\ddot{\phi}_{j\pm1} - C_B\ddot{\phi}_{j\pm N}\right] = 2E_J\left(\frac{1}{2}(\phi_{j+1} + \phi_{j-1}) - \phi_j\right) \tag{1.14}$$

The dispersion relation for this system can be found easily, seeking solutions of the form $\phi_j = exp(i(k.ja - \omega t))$

$$\frac{1}{2}\left(\frac{\phi_0}{2\pi}\right)^2 \omega^2 \exp(i(kja - \omega t))\left[(C_g + 2C_J + C_B) - C_J e^{\pm ika} - C_B e^{\pm iNka}\right] = 2E_J \exp(i(kja - \omega t))\left(\frac{1}{2}(e^{ika} + e^{-ika}) - 1\right) \tag{1.15}$$

Simplifying, we get

$$\omega^2 = \frac{-4E_J \sin^2\left(\frac{ka}{2}\right)}{\frac{1}{2}\left(\frac{\phi_0}{2\pi}\right)^2 \left[C_g + 4C_J\sin^2(\frac{ka}{2}) + C_B(1 - 2\cos(Nka))\right]} \tag{1.16}$$

Chapter 2

Obtaining the inverse of the capacitance matrix

2.1 Solving the Toeplitz matrix

A Toeplitz matrix is a matrix where all the elements in a particular diagonal has the same value.

$$\begin{bmatrix} a & b & c & d & e \\ f & a & b & c & d \\ g & f & a & b & c \\ h & g & f & a & b \\ i & h & g & f & a \end{bmatrix} \longrightarrow \text{An example of a Toeplitz matrix} \tag{2.1}$$

We can now observe that the matrix C in is an example of a Toeplitz matrix. There are a lot of ways we can solve a Topelitz matrix, examples include Schur's algorithm, Levinson's algorithm and LU decomposition by the Bareiss algorithm. We shall look at one method to solve this system. The matrix we have is

$$\begin{bmatrix} 2C_J+C_B+C_g & -C_J & 0 & \ldots\ldots\ldots & -C_B & 0 & \ldots\ldots\ldots\ldots & 0 \\ -C_J & 2C_J+C_B+C_g & -C_J & \ldots\ldots\ldots & 0 & -C_B & \ldots\ldots\ldots & 0 \\ 0 & -C_J & 2C_J+C_B+C_g & \ldots\ldots\ldots & 0 & 0 & \ldots\ldots\ldots\ldots & 0 \\ \ldots. & \ldots. & \ldots. & \ldots\ldots. & \ldots\ldots & \ldots\ldots & \ldots\ldots. & \ldots\ldots \\ 0 & 0 & 0 & \ldots\ldots\ldots & C_J & 0 & \ldots\ldots. & -C_B \\ -C_B & 0 & 0 & \ldots\ldots\ldots & 2C_J+C_B+C_g & C_J & \ldots\ldots & 0 \\ 0 & -C_B & 0 & \ldots\ldots\ldots & C_J & 2C_J+C_B+C_g & \ldots\ldots. & 0 \\ \ldots. & \ldots. & \ldots. & \ldots\ldots. & \ldots\ldots & \ldots\ldots & \ldots\ldots. & \ldots\ldots \\ 0 & 0 & 0 & \ldots\ldots & 0 & 0 & \ldots.. & 2C_J+C_B+C_g \end{bmatrix}$$

The Levinson Algorithm

We have H**v**=**y**, where H is the matrix to be solved, **v** are the variables and **y** are the values on the right hand side. Here we assume H is a symmetric Toeplitz matrix. H= $\begin{bmatrix} h_1 & h_2 & h_3 & \cdots \\ h_2 & h_1 & h_2 & \cdots \\ h_3 & \ddots & & \cdots \\ \cdots & \cdots & \cdots & \cdots \end{bmatrix}$ $=H^T$

We introduce a "counter-identity" matrix J defined as, J= $\begin{bmatrix} 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 1 & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 1 & 0 & \cdots & 0 & 1 \\ 1 & 0 & 0 & \cdots & 1 & 0 \end{bmatrix}$.

J is also called the exchange matrix as it can be used to exchange rows or columns. For example, if $x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, then $Jx = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$

So the action JA reverses the columns of A and AJ reverses the rows of A. Applying J to the symmetric Toeplitz matrix,

$$JH = \begin{bmatrix} h_{N-1} & h_{N-2} & \cdots & \cdots & h_0 \\ h_{N-2} & h_{N-1} & \cdots & \cdots & \vdots \\ \ddots & \ddots & \ddots & \cdots & \vdots \\ h_o & \cdots & \cdots & \cdots & h_{N-1} \end{bmatrix}$$

$$JHJ = \begin{bmatrix} h_1 & h_2 & h_3 & \cdots \\ h_2 & h_1 & h_2 & \cdots \\ h_3 & \ddots & & \cdots \\ \cdots & \cdots & \cdots & \cdots \end{bmatrix} = H$$

$$\begin{aligned} JHJ = H \quad &\implies JH = HJ \\ H^{-1} = (JHJ) = J^{-1}H^{-1}J^{-1} \quad &\implies H^{-1}J = J^{-1}H \end{aligned}$$

Writing in the Block diagonal form,

$$\left[\begin{array}{cccc|c} h_0 & h_1 & h_2 & \cdots & h_{N-1} \\ h_1 & h_0 & h_1 & \cdots & h_{N-2} \\ h_2 & h_1 & h_0 & \cdots & h_{N-1} \\ \vdots & \ddots & \ddots & \ddots & \cdots \\ \cdots & \cdots & \cdots & \cdots & h_1 \\ \hline h_{N-1} & h_{N-2} & h_{N-3} & \cdots h_1 & h_0 \end{array}\right] \left[\begin{array}{c} v_1 \\ v_2 \\ v_3 \\ \vdots \\ \vdots \\ \hline v_N \end{array}\right] = \left[\begin{array}{c} y_1 \\ y_2 \\ y_3 \\ \vdots \\ \vdots \\ \hline y_N \end{array}\right]$$

$$\left[\begin{array}{c|c} H_{N-1} & Jh_{N-1} \\ \hline (Jh_{N-1})^T & h_0 \end{array}\right] \left[\begin{array}{c} w \\ \hline \alpha \end{array}\right] = \left[\begin{array}{c} \mathbf{Y}_{N-1} \\ \hline y_N \end{array}\right] \tag{2.2}$$

$$\implies H_{N-1}w + \alpha J h_{N-1} = \mathbf{Y}_{N-1} \tag{2.3}$$

$$h_{N-1}^T J w + \alpha h_0 = y_N \tag{2.4}$$

From 2.3,

$$w = H_{N-1}^{-1}\left(\mathbf{Y}_{N-1} - \alpha J h_{N-1}\right) \tag{2.5}$$

If we know

$$X_{N-1} = H_{N-1}^{-1}\mathbf{Y}_{N-1}; V_{N-1} = H_{N-1}^{-1}h_{N-1}$$

then $w = X_{N-1} - \alpha J v_{N-1}$

$$\implies \alpha = \frac{y_N - h_{N-1}^T J X_{N-1}}{h_0 - h_{N-1}^T v_{N-1}} \tag{2.6}$$

So if the following are given,

$$\begin{aligned} H_{N-1}X_{N-1} &= \mathbf{Y}_{N-1} \\ H_{N-1}v_{N-1} &= h_{N-1} \end{aligned}$$

we can compute the solution to

$$H_N X_N = \mathbf{Y}_N; X_N = \begin{bmatrix} w \\ \alpha \end{bmatrix} = \begin{bmatrix} X_{N-1} - \alpha J v_{N-1} \\ \alpha \end{bmatrix} \tag{2.7}$$

Hence we have obtained a recursion with which we can approach this problem.

2.2 Inverse of the Capacitance Matrix

Doing a row transformation, the matrix becomes

$$\begin{bmatrix} 2C_J + C_g & -C_J & 0 & \ldots\ldots\ldots & 2C_J + C_g & -C_J & \ldots\ldots\ldots. & 0 \\ -C_J & 2C_J + C_g & -C_J & \ldots\ldots\ldots & -C_J & 2C_J + C_g & \ldots\ldots.. & 0 \\ 0 & -C_J & 2C_J + C_g & \ldots\ldots\ldots & 0 & 0 & \ldots\ldots\ldots. & 0 \\ \ldots. & \ldots. & \ldots. & \ldots\ldots. & \ldots\ldots & \ldots\ldots & \ldots\ldots. & \ldots\ldots \\ 0 & 0 & 0 & \ldots\ldots.. & -C_J & 0 & \ldots\ldots. & 2C_J + C_g \\ -C_B & 0 & 0 & \ldots\ldots\ldots & 2C_J + C_B + C_g & -C_J & \ldots\ldots & 0 \\ 0 & -C_B & 0 & \ldots\ldots\ldots & -C_J & 2C_J + C_B + C_g & \ldots\ldots. & 0 \\ \ldots. & \ldots. & \ldots. & \ldots\ldots. & \ldots\ldots & \ldots\ldots & \ldots\ldots. & \ldots\ldots \\ 0 & 0 & 0 & \ldots\ldots & 0 & 0 & \ldots.. & 2C_J + C_B + C_g \end{bmatrix}$$

$$\begin{bmatrix} 2C_J+C_g & -C_J & 0 & .... & 0 & 0 & .... & 0 \\ -C_J & 2C_J+C_g & -C_J & .... & 0 & 0 & .... & 0 \\ 0 & -C_J & 2C_J+C_g & .... & 0 & 0 & .... & 0 \\ .... & .... & .... & .... & ...... & ...... & .... & ...... \\ 0 & 0 & 0 & .... & 0 & 0 & .... & 0 \\ -C_B & 0 & 0 & .... & 2C_J+2C_B+C_g & -C_J & ...... & 0 \\ 0 & -C_B & 0 & .... & -C_J & 2C_J+2C_B+C_g & .... & 0 \\ .... & .... & .... & .... & ...... & ...... & .... & ...... \\ 0 & 0 & 0 & ...... & 0 & 0 & ..... & 2C_J+2C_B+C_g \end{bmatrix}$$

Doing another set of elimination gives us the block matrix

$$\widetilde{C} = \begin{bmatrix} C_+ & 0 \\ 0 & C_- \end{bmatrix}$$

Taking $C_+ = C_g + 2C_J$ and $C_- = C_g + 2C_J + 2C_B$ We finally obtain

$$C_\pm = \frac{C_J}{2} \begin{bmatrix} x_\pm & -1 & 0 & \cdots & \mp 1 \\ -1 & x_\pm & -1 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \mp 1 & 0 & \cdots & -1 & x_\pm \end{bmatrix}$$

where $x_\pm = C_\pm / C_J$. Applying the algorithm developed above on this system, or by directly applying results from [3], we finally get the equation obtained by Koch.

$$\left(C_\pm^{-1}\right)_{j,k} = \frac{U_{k-j-1}(x_\pm/2) - U_{N-k+j-1}(x_\pm/2)}{C_J\left[1 - T_N(x_\pm/2)\right]} \tag{2.8}$$

Where T_n and U_n are Chebyshev polynomials of first and second kind. We take the charging energy matrix

$$\left(E_C^\pm\right) = \frac{e^2}{2}\left(C_\pm^{-1}\right)_{j,k} \tag{2.9}$$

2.3 Exciton, Agiton variables and deriving the Hamiltonian of the system

We define the Exciton (-) and Agiton (+) variables to be,

$$\phi_j^\pm = \phi_j \pm \phi_{N+j} \tag{2.10}$$

and the momentum operator as

$$p_j^\pm = \frac{p_j \pm p_{N+j}}{2} \tag{2.11}$$

where the above operators follow the canonical rule

$$\left[\phi_j^\sigma, p_k^\tau\right] = i\delta_{jk}\delta_{\sigma\tau} \qquad\qquad \sigma,\tau = \pm \tag{2.12}$$

This definition follows from the the matrix transformations we performed to reduce it to the block diagonal form. The momentum operators follow a simple constraint that for each rung j, the momentum must be either both integer or both half integers.
We have the Hamiltonian after changing the variables and applying trigonometric identities.

$$\begin{aligned} H = &\sum_{\sigma=\pm}\sum_{i,j=1}^{N} 4(E_C^\sigma)_{i,j}\left(p_i^\sigma - n_{gi}^\sigma\right)\left(p_j^\sigma - n_{gj}^\sigma\right) \\ &- \sum_{j=1}^{N-1} 2E_J cos\left(\frac{1}{2}\left[\phi_{j+1}^+ - \phi_j^+ - \phi^{ext}/N\right]\right) cos\left(\frac{1}{2}\left[\phi_{j+1}^- - \phi_j^-\right]\right) \\ &- 2E_J cos\left(\frac{1}{2}\left[\phi_1^+ - \phi_N^+ - \phi^{ext}/N\right]\right) cos\left(\frac{1}{2}\left[\phi_1^- + \phi_N^-\right]\right) \end{aligned} \tag{2.13}$$

Chapter 3

Obtaining the wave equations of the system

3.1 Another try on the dispersion relation

In the exciton subspace, we obtain the Lagrangian around $0, \pi$ as [1],

$$\mathcal{L}^{0,\pi}_{eff} = \frac{1}{2}\left(\frac{\phi_0}{2\pi}\right)^2 \sum_{i,j=1}^{N} \dot{\phi_i^-}(C_-)_{ij}\dot{\phi_j^-} + \sum_{j=1} \frac{J}{2}\left(\phi^-_{j+1} - \phi^-_j\right)^2 + \frac{J}{2}(\phi^-_1 + \phi^-_N) \tag{3.1}$$

Obtaining the Euler-Lagrange equations, we get

$$\frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial \dot{\phi_j^-}}\right) = \frac{1}{2}\left(\frac{\phi_0}{2\pi}\right)^2 \sum_{i=1}^{2N} \ddot{\phi_i^-}(C_-)_{ij} \text{ ; } \frac{\partial \mathcal{L}}{\partial \phi_j^-} = -J(\phi^-_{j+1} - \phi^-_j) + J(\phi^-_j - \phi^-_{j-1})$$

$$\frac{1}{2}\left(\frac{\phi_0}{2\pi}\right)^2 \sum_{i=1}^{2N} \ddot{\phi_i^-}(C_-)_{ij} = -J(\phi^-_{j+1} - 2\phi^-_j + \phi^-_{j-1}) \tag{3.2}$$

Doing the same procedure as section 1.4 and then setting a appropriately, we get the relation obtained by Weiss et.al in [1]

$$\begin{aligned}\omega_k &= \frac{2\pi}{\Phi_0}\sqrt{\frac{4J\sin^2\left[(2k-1)\pi/2N\right]}{C_B + C_g/2 + 2C_J\sin^2\left[(2k-1)\pi/2N\right]}} \\ &= \frac{2\pi}{\Phi_0}\sqrt{\frac{4J}{C_B}}\left|\sin\frac{(2k-1)\pi}{2N}\right| + \mathcal{O}\left(\tfrac{C_g}{C_B}, \tfrac{C_J}{C_B}\right).\end{aligned} \tag{3.3}$$

3.2 Wave equation in the exciton subspace

If we perform the Schrieffer-Wolff transformation of 4.10, and eliminate the high energy subspace of agitons, we get the Hamiltonian as done in [1]

$$H = \sum_{i,j=1}^{N} 4(E^-_c)_{i,j}(p^-_i - n_{gi})(p^-_j - n_{gj}) - \sum_{j=1}^{N-1} J_j cos\left(\phi^-_{j+1} - \phi^-_j\right) - J_N cos\left(\phi^-_1 + \phi^-_N\right) \tag{3.4}$$

For protected regime and vanishing offset charges

$$J \approx \frac{E_J^2}{2E_{C_J}}$$

Taking the continuum limit and Taylor expanding cosine around 0 or π.

$$H = \int_{x,y=1}^{N} 4dxdy(E^-_c(x,y))(p^-(x))(p^-(y)) - \int_{y=1}^{N-1} dyJ\left[1 - \frac{1}{2}\left(\phi^-(x+1) - \phi^-(x)\right)^2\right] - J_N cos\left(\phi^-_1 + \phi^-_N\right) \tag{3.5}$$

For excitons, the charging energies are very short-ranged with off-diagonal elements decreasing rapidly in powers of CJ/CB . The primarily relevant entries of E^-_C are along the diagonal and first off-diagonals

$$\begin{aligned}\left(\mathsf{E}^-_\mathsf{C}\right)_{j,j} &= \frac{e^2}{2C_B} + \mathcal{O}\left(\tfrac{C_J}{C_B}, \tfrac{C_g}{C_B}, \tfrac{1}{N}\right), \\ \left(\mathsf{E}^-_\mathsf{C}\right)_{j,j+1} &= \frac{e^2 C_J}{4C_B^2} + \mathcal{O}\left(\tfrac{C_J}{C_B}, \tfrac{C_g}{C_B}, \tfrac{1}{N}\right), \\ \left(\mathsf{E}^-_\mathsf{C}\right)_{1,N} &= -\frac{e^2 C_J}{4C_B^2} + \mathcal{O}\left(\tfrac{C_J}{C_B}, \tfrac{C_g}{C_B}, \tfrac{1}{N}\right).\end{aligned} \tag{3.6}$$

We can use the approximation derived

$$E_c^- = \frac{e^2}{2C_B}\delta(x,y) + \frac{e^2 C_J}{2C_B}\delta(x+1,y) - \frac{e^2 C_J}{2C_B}\delta(1,N)$$

to get

$$\begin{aligned} H = &\int_{x,y=1}^{N} 4dxdy\left[\frac{e^2}{2C_B}\delta(x,y) + \frac{e^2 C_J}{2C_B}\delta(x+1,y) - \frac{e^2 C_J}{2C_B}\delta(1,N)\right](p^-(x))(p^-(y)) \\ &- \int_{x=1}^{N-1} dyJ\left[1 - \frac{1}{2}\left(\phi_{j+1}^- - \phi_j^-\right)^2\right] - J_N cos\left(\phi_1^- + \phi_N^-\right) \end{aligned} \tag{3.7}$$

which simplifies to

$$\begin{aligned} H = &\int_{x,y=1}^{N} 4dxdy\left[\frac{e^2}{2C_B}\delta(x,y) + \frac{e^2 C_J}{2C_B}\delta(x\pm1,y) - \frac{e^2 C_J}{2C_B}\delta(1,N)\right](p^-(x))(p^-(y)) \\ &- \int_{y=1}^{N-1} dyJ\left[1 - \frac{1}{2}\left(\phi^-(x+1) - \phi^-(x)\right)^2\right] - J_N cos\left(\phi_1^- + \phi_N^-\right) \end{aligned} \tag{3.8}$$

$$\begin{aligned} H = &\int_{x=1}^{N} 4dx\left[\frac{e^2}{2C_B}(p^-(x))^2 + \frac{e^2 C_J}{2C_B}\left(p^-(x)p^-(x\pm1)\right) - \frac{e^2 C_J}{2C_B}(p^-(1))(p^-(N))\right] \\ &- \int_{y=1}^{N-1} dyJ\left[1 - \frac{1}{2}a^2\left(\frac{\partial\phi^-(x)}{\partial x}\right)^2\right] - J_N cos\left(\phi_1^- + \phi_N^-\right) \end{aligned} \tag{3.9}$$

where we have taken $\left(\phi^-(x+1) - \phi^-(x)\right)/a = \frac{\partial\phi^-(x)}{\partial x}$.
The equation 3.9 can be used to obtain our wave equation, by applying Hamilton's equations of motion

$$\frac{\partial H}{\partial \phi} = -\dot{p} \qquad \frac{\partial H}{\partial p} = \dot{\phi}$$

Taking the second equation

$$\frac{\partial H}{\partial p^-(x)} = 4\left[\frac{e^2}{2C_B}2(p^-(x)) + \frac{e^2 C_J}{2C_B}p^-(x\pm1)\right] = \dot{\phi} \tag{3.10}$$

we get

$$p^-(x) = \frac{C_B}{4e^2}\left[\dot{\phi}^-(x) - 2e^2\frac{C_J}{C_B}p^-(x\pm1)\right] \tag{3.11}$$

Taking the first Hamilton's equation

$$\frac{\partial H}{\partial \phi^-(x)} = -J\frac{1}{2}a^2 2\frac{\partial\phi^-(x)}{\partial x}\frac{\partial}{\partial\phi^-(x)}\left(\frac{\partial\phi^-(x)}{\partial x}\right) = -\dot{p} \tag{3.12}$$

by inverse function theorem

$$\frac{\partial}{\partial x}\left(\frac{\partial x}{\partial y}\right) = \frac{\frac{\partial^2 x}{\partial y^2}}{\frac{\partial x}{\partial y}}$$

Applying this,

$$\dot{p} = Ja^2\left(\frac{\partial^2\phi^-(x)}{\partial x^2}\right) \tag{3.13}$$

$$\frac{\partial}{\partial t}\frac{C_B}{4e^2}\left[\dot{\phi} - 2e^2\frac{C_J}{C_B}p^-(x\pm1)\right] = Ja^2\left(\frac{\partial^2\phi^-(x)}{\partial x^2}\right) \tag{3.14}$$

$$\frac{C_B}{4e^2}\left[\ddot{\phi} - 2e^2\frac{C_J}{C_B}\dot{p}^-(x\pm1)\right] = Ja^2\left(\frac{\partial^2\phi^-(x)}{\partial x^2}\right) \tag{3.15}$$

3.15 is the wave equation of the exciton subspace. If we make $C_J << C_B$ We get the standard wave equation

$$\ddot{\phi}^-(x) = \frac{4e^2}{C_B}Ja^2\left(\frac{\partial^2\phi^-(x)}{\partial x^2}\right) \tag{3.16}$$

3.3 Wave equations in the general case

We now try to extract the general wave equation by using both exciton and agiton variables. We have the Hamiltonian

$$H = \sum_{\sigma=\pm}\sum_{i,j=1}^{N} 4(E_C^\sigma)_{i,j}\left(p_i^\sigma - n_{gi}^\sigma\right)\left(p_j^\sigma - n_{gj}^\sigma\right)$$
$$- \sum_{j=1}^{N-1} 2E_J cos\left(\frac{1}{2}\left[\phi_{j+1}^+ - \phi_j^+ - \phi^{ext}/N\right]\right) cos\left(\frac{1}{2}\left[\phi_{j+1}^- - \phi_j^-\right]\right)$$
$$- 2E_J cos\left(\frac{1}{2}\left[\phi_1^+ - \phi_N^+ - \phi^{ext}/N\right]\right) cos\left(\frac{1}{2}\left[\phi_1^- + \phi_N^-\right]\right) \tag{3.17}$$

Doing the same procedure as above, we get

$$H = \int_{x,y=1}^{N} 4dxdy\left[(E_c^-(x,y))(p^-(x)p^-(y)) + (E_c^+(x,y))(p^+(x)p^+(y))\right]$$
$$- \int_{y=1}^{N-1} dyE_J\left[1 - \frac{1}{8}a^2\left(\frac{\partial\phi^+(x)}{\partial x}\right)^2\right]\left[1 - \frac{1}{8}a^2\left(\frac{\partial\phi^-(x)}{\partial x}\right)^2\right]$$
$$- 2E_J cos\left(\frac{1}{2}\left[\phi_1^+ - \phi_N^+ - \phi^{ext}/N\right]\right) cos\left(\frac{1}{2}\left[\phi_1^- + \phi_N^-\right]\right) \tag{3.18}$$

Now, Agiton charging energies are long ranged, with maximum entries in the diagonal region and decreasing monotonously to the N/2th diagonal. We shall consider the diagonal and the first off diagonal for this exercise

$$E_c^+ = \frac{e^2}{NC_g}\left(1 + \frac{C_g(N^2-1)}{12C_J}\right)\delta(x,y) + \frac{e^2}{NC_g}\left(1 + \frac{C_g(N^2-6N+5)}{12C_J}\right)\delta(x\pm 1, y)$$

Doing the similar treatment gives us the Hamiltonian as

$$H = \int_{x=1}^{N} 4dx\left[\frac{e^2}{2C_B}(p^-(x))^2 + \frac{e^2C_J}{2C_B}\left(p^-(x)p^-(x\pm 1)\right) - \frac{e^2C_J}{2C_B}(p^-(1))(p^-(N))\right]$$
$$+ \int_{x=1}^{N} 4dx\left[\frac{e^2}{NC_g}\left(1 + \frac{C_g(N^2-1)}{12C_J}\right)(p^+(x))^2 + \frac{e^2}{NC_g}\left(1 + \frac{C_g(N^2-6N+5)}{12C_J}\right)\left(p^+(x)p^+(x\pm 1)\right)\right]$$
$$- \int_{y=1}^{N-1} dyE_J\left[1 - \frac{1}{8}a^2\left(\frac{\partial\phi^+(x)}{\partial x}\right)^2\right]\left[1 - \frac{1}{8}a^2\left(\frac{\partial\phi^-(x)}{\partial x}\right)^2\right]$$
$$- 2E_J cos\left(\frac{1}{2}\left[\phi_1^+ - \phi_N^+ - \phi^{ext}/N\right]\right) cos\left(\frac{1}{2}\left[\phi_1^- + \phi_N^-\right]\right) \tag{3.19}$$

We again invoke Hamilton's equation of motion

$$\frac{\partial H}{\partial \phi} = -\dot{p} \qquad \frac{\partial H}{\partial p} = \dot{\phi}$$

This time, we might get two wave equations, each for ϕ^- and ϕ^+. First performing for ϕ^- we get

$$\frac{\partial H}{\partial p^-(x)} = 4\left[\frac{e^2}{2C_B}2(p^-(x)) + \frac{e^2C_J}{2C_B}\left[p^-(x+1) + p^-(x-1)\right]\right] = \dot{\phi} \tag{3.20}$$

$$p^-(x) = \frac{C_B}{e^2}\left[\frac{\dot{\phi}}{4} - \frac{e^2C_J}{2C_B}\left(p^-(x+1) + p^-(x+1)\right)\right] \tag{3.21}$$

$$\frac{\partial H}{\partial \phi^-(x)} = -\dot{p}^-(x) = E_J\left[1 - \frac{1}{8}a^2\left(\frac{\partial\phi^+(x)}{\partial x}\right)\right]\left[\frac{1}{4}a^2\left(\frac{\partial^2\phi^-(x)}{\partial x^2}\right)\right] \tag{3.22}$$

$$\frac{C_B}{e^2}\left[\frac{\ddot{\phi}}{4} - \frac{e^2C_J}{2C_B}\left(\dot{p}^-(x+1) + \dot{p}^-(x+1)\right)\right] = E_J\left[1 - \frac{1}{8}a^2\left(\frac{\partial\phi^+(x)}{\partial x}\right)^2\right]\left[\frac{1}{4}a^2\left(\frac{\partial^2\phi^-(x)}{\partial x^2}\right)\right] \tag{3.23}$$

3.23 is the wave equation for ϕ^- variable, the ϕ^+ term acts as a perturbation in this case. Doing the same, but for ϕ_+

$$\frac{\partial H}{\partial p^+(x)} = 4\left[\frac{e^2}{NC_g}\left(1 + \frac{C_g(N^2-1)}{12C_J}\right)2(p^+(x)) + \frac{e^2}{NC_g}\left(1 + \frac{C_g(N^2-6N+5)}{12C_J}\right)\left[p^+(x+1) + p^+(x-1)\right]\right] = \dot{\phi^+} \tag{3.24}$$

$$p^+(x) = \frac{NC_g}{2e^2}\frac{1}{\left(1+\frac{C_g(N^2-1)}{12C_J}\right)}\left[\frac{\dot{\phi}^+}{4} - \frac{e^2}{NC_g}\left(1+\frac{C_g(N^2-6N+5)}{12C_J}\right)(p^-(x+1)+p^-(x+1))\right] \tag{3.25}$$

$$\frac{\partial H}{\partial \phi^+(x)} = -\dot{p}^+(x) = E_J\left[1-\frac{1}{8}a^2\left(\frac{\partial \phi^-(x)}{\partial x}\right)\right]\left[\frac{1}{4}a^2\left(\frac{\partial^2\phi^+(x)}{\partial x^2}\right)\right] \tag{3.26}$$

$$\begin{aligned}&\frac{NC_g}{2e^2}\frac{1}{\left(1+\frac{C_g(N^2-1)}{12C_J}\right)}\left[\frac{\ddot{\phi}^+}{4} - \frac{e^2}{NC_g}\left(1+\frac{C_g(N^2-6N+5)}{12C_J}\right)(\dot{p}^+(x+1)+\dot{p}^+(x-1))\right]\\ &= E_J\left[1-\frac{1}{8}a^2\left(\frac{\partial \phi^-(x)}{\partial x}\right)^2\right]\left[\frac{1}{4}a^2\left(\frac{\partial^2\phi^+(x)}{\partial x^2}\right)\right]\end{aligned} \tag{3.27}$$

We have the coupled equations,

$$\frac{C_B}{e^2}\left[\frac{\ddot{\phi}^-}{4} - \frac{e^2C_J}{2C_B}(\dot{p}^-(x+1)+\dot{p}^-(x+1))\right] = E_J\left[1-\frac{1}{8}a^2\left(\frac{\partial \phi^+(x)}{\partial x}\right)\right]\left[\frac{1}{4}a^2\left(\frac{\partial^2\phi^-(x)}{\partial x^2}\right)\right]$$

$$\begin{aligned}&\frac{NC_g}{2e^2}\frac{1}{\left(1+\frac{C_g(N^2-1)}{12C_J}\right)}\left[\frac{\ddot{\phi}^+}{4} - \frac{e^2}{NC_g}\left(1+\frac{C_g(N^2-6N+5)}{12C_J}\right)(\dot{p}^+(x+1)+\dot{p}^+(x-1))\right]\\ &= E_J\left[1-\frac{1}{8}a^2\left(\frac{\partial \phi^-(x)}{\partial x}\right)\right]\left[\frac{1}{4}a^2\left(\frac{\partial^2\phi^+(x)}{\partial x^2}\right)\right]\end{aligned}$$

Setting the momentum terms to 0 as the leading coefficients are negligible,

$$\frac{C_B}{e^2}\left[\frac{\ddot{\phi}^-}{4}\right] = E_J\left[1-\frac{1}{8}a^2\left(\frac{\partial \phi^+(x)}{\partial x}\right)^2\right]\left[\frac{1}{4}a^2\left(\frac{\partial^2\phi^-(x)}{\partial x^2}\right)\right] \tag{3.28}$$

$$\frac{NC_g}{2e^2}\frac{1}{\left(1+\frac{C_g(N^2-1)}{12C_J}\right)}\left[\frac{\ddot{\phi}^+}{4}\right] = E_J\left[1-\frac{1}{8}a^2\left(\frac{\partial \phi^-(x)}{\partial x}\right)^2\right]\left[\frac{1}{4}a^2\left(\frac{\partial^2\phi^+(x)}{\partial x^2}\right)\right] \tag{3.29}$$

Setting

$$\frac{C_B}{e^2} = \alpha$$
$$\frac{NC_g}{2e^2}\frac{1}{\left(1+\frac{C_g(N^2-1)}{12C_J}\right)} = \beta$$
$$E_Ja^2 = E$$

to obtain,

$$\alpha\left[\ddot{\phi}^-\right] = E\left[1-\frac{1}{8}a^2\left(\frac{\partial \phi^+(x)}{\partial x}\right)^2\right]\left(\frac{\partial^2\phi^-(x)}{\partial x^2}\right) \tag{3.30}$$

$$\beta\left[\ddot{\phi}^+\right] = E\left[1-\frac{1}{8}a^2\left(\frac{\partial \phi^-(x)}{\partial x}\right)^2\right]\left(\frac{\partial^2\phi^+(x)}{\partial x^2}\right) \tag{3.31}$$

This is yet to be solved.

3.4 Solution in the exciton subspace:

We first take the case of the exciton subspace

$$\frac{\partial^2\phi^-(x)}{\partial t^2} = \frac{4e^2}{C_B}Ja^2\left(\frac{\partial^2\phi^-(x)}{\partial x^2}\right) \tag{3.32}$$

The speed of the wave is given by

$$c^2 = \frac{4e^2}{C_B}Ja^2 = \frac{4e^2}{C_B}\frac{E_J^2}{2E_{C_J}}a^2 \tag{3.33}$$

The solution of the wave equation should be of the form

$$\phi^-(x) = \phi(x) - \phi(x+N) = f(x+ct) - g(x-ct) \tag{3.34}$$

Where f and g are oscillatory functions. Using the boundary condition, that $c(\phi(x)) = -c(\phi(x+N))$.

$$\phi^-(x) = \phi(x) - \phi(x+N) = f(x+ct) - f(x-ct) \tag{3.35}$$

Chapter 4

Conclusions and results

- We have analyzed the Current mirror qubit, the key results are presented below
  - The Lagrangian of the system in the normal subspace is given by
$$\mathcal{L} = \frac{1}{2}\left(\frac{\phi_0}{2\pi}\right)^2 \sum_{i,j=1}^{2N} \dot{\phi}_i C_{ij} \dot{\phi}_j + \hbar \sum_{j=1}^{2N} \dot{\phi}_j n_{gj} + \sum_{j=1}^{2N} E_J \cos\left(\phi_{j+1} - \phi_j - \frac{\phi_{ext}}{2N}\right)$$
  - The equation of motion of the system is given by
$$\frac{1}{2}\left(\frac{\phi_0}{2\pi}\right)^2 \left[(C_g + 2C_J + C_B)\ddot{\phi}_j - C_J\ddot{\phi}_{j\pm1} - C_B\ddot{\phi}_{j\pm N}\right] + \hbar\dot{n}_{gj} = E_J \sin\left(\phi_{j+1} - \phi_j + \frac{\phi_{ext}}{2N}\right) - E_J \sin\left(\phi_j - \phi_{j-1} + \frac{\phi_{ext}}{2N}\right) \tag{4.1}$$
  - The Hamiltonian of the system in normal subspace is given by
$$H = \sum_{i,j=1}^{2N} 4E_{c_{i,j}} (p_i - n_{gi})(p_j - n_{gj}) - 2\sum_{j=1}^{2N} E_J \cos\left(\phi_{j+1} - \phi_j - \frac{\phi_{ext}}{2N}\right) \tag{4.2}$$
  - The Continuum Hamiltonian is given by
$$H = \int_{y=1}^{2N} dy \left[\left(4\int_{x=1}^{2N} dx.E_c(x,y)\left(\frac{d\phi(x)}{dt}\frac{d\phi(y)}{dt}\right)dx\right) - 2E_J a^2 \left(\frac{d\phi(y)}{dx}\right)^2\right] \tag{4.3}$$
  - The dispersion relation is given by
$$\omega^2 = \frac{-4E_J \sin^2\left(\frac{ka}{2}\right)}{\frac{1}{2}\left(\frac{\phi_0}{2\pi}\right)^2 \left[C_g + 4C_J \sin^2(\frac{ka}{2}) + C_B(1 - 2\cos(Nka))\right]} \tag{4.4}$$
- We devised an algorithm to find the inverse of the capacitance matrix
- The inverse of the capacitance matrix is given by,
$$\left(C_{\pm}^{-1}\right)_{j,k} = \frac{U_{k-j-1}(x_{\pm}/2) - U_{N-k+j-1}(x_{\pm}/2)}{C_J\left[1 - T_N(x_{\pm}/2)\right]} \tag{4.5}$$
  Where T_n and U_n are Chebyshev polynomials of first and second kind.
- We obtained the charging energy matrix
$$(E_C^{\pm}) = \frac{e^2}{2}\left(C_{\pm}^{-1}\right)_{j,k} \tag{4.6}$$
- We defined the Exciton (-) and Agiton (+) variables, as an outcome to the matrix transformations, to be,
$$\phi_j^{\pm} = \phi_j \pm \phi_{N+j} \tag{4.7}$$
  And the momentum operator as
$$p_j^{\pm} = \frac{p_j \pm p_{N+j}}{2} \tag{4.8}$$
  Where the above operators follow the canonical rule
$$\left[\phi_j^{\sigma}, p_k^{\tau}\right] = i\delta_{jk}\delta_{\sigma\tau} \qquad \sigma, \tau = \pm \tag{4.9}$$

- The Hamiltonian after effecting the changes turns out to be

$$
\begin{aligned}
H = &\sum_{\sigma=\pm}\sum_{i,j=1}^{N} 4(E_C^\sigma)_{i,j}\left(p_i^\sigma - n_{gi}^\sigma\right)\left(p_j^\sigma - n_{gj}^\sigma\right) \\
&- \sum_{j=1}^{N-1} 2E_J cos\left(\frac{1}{2}\left[\phi_{j+1}^+ - \phi_j^+ - \phi^{ext}/N\right]\right) cos\left(\frac{1}{2}\left[\phi_{j+1}^- - \phi_j^-\right]\right) \\
&- 2E_J cos\left(\frac{1}{2}\left[\phi_1^+ - \phi_N^+ - \phi^{ext}/N\right]\right) cos\left(\frac{1}{2}\left[\phi_1^- + \phi_N^-\right]\right)
\end{aligned}
\tag{4.10}
$$

- The Lagrangian near 0,π turns out to be

$$
\mathcal{L}_{eff}^{0,\pi} = \frac{1}{2}\left(\frac{\phi_0}{2\pi}\right)^2 \sum_{i,j=1}^{N} \dot{\phi_i^-}(C_-)_{ij}\dot{\phi_j^-} + \sum_{j=1}\frac{J}{2}\left(\phi_{j+1}^- - \phi_j^-\right)^2 + \frac{J}{2}(\phi_1^- + \phi_N^-)
\tag{4.11}
$$

- The dispersion relation turns out to be,

$$
\begin{aligned}
\omega_k &= \frac{2\pi}{\Phi_0}\sqrt{\frac{4J\sin^2\left[(2k-1)\pi/2N\right]}{C_B + C_g/2 + 2C_J\sin^2\left[(2k-1)\pi/2N\right]}} \\
&= \frac{2\pi}{\Phi_0}\sqrt{\frac{4J}{C_B}}\left|\sin\frac{(2k-1)\pi}{2N}\right| + \mathcal{O}\left(\tfrac{C_g}{C_B},\tfrac{C_J}{C_B}\right).
\end{aligned}
\tag{4.12}
$$

- The Hamilton in exciton space after the Schrieffer-Wolff transformation turns out to be

$$
H = \sum_{i,j=1}^{N} 4(E_c^-)_{i,j}(p_i^- - n_{gi})(p_j^- - n_{gj}) - \sum_{j=1}^{N-1} J_j cos\left(\phi_{j+1}^- - \phi_j^-\right) - J_N cos\left(\phi_1^- + \phi_N^-\right)
\tag{4.13}
$$

- The wave equation in the exciton case is given by

$$
\ddot{\phi}^-(x) = \frac{4e^2}{C_B}Ja^2\left(\frac{\partial^2\phi^-(x)}{\partial x^2}\right)
\tag{4.14}
$$

- The solution of the wave equation turns out to be

$$
\phi^-(x) = \phi(x) - \phi(x+N) = f(x+ct) - f(x-ct)
\tag{4.15}
$$

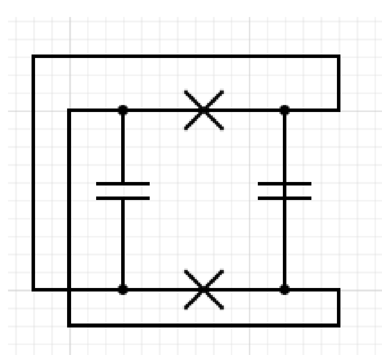
  This means that there are two waves running through the two stripes but in opposite direction, in such a way that the canonical rule is obeyed.

- The coupled wave equation in the general subspace is given by

$$
\frac{C_B}{e^2}\left[\frac{\ddot{\phi}^-}{4} - \frac{e^2C_J}{2C_B}\left(\dot{p}^-(x+1) + \dot{p}^-(x+1)\right)\right] = E_J\left[1 - \frac{1}{8}a^2\left(\frac{\partial\phi^+(x)}{\partial x}\right)\right]\left[\frac{1}{4}a^2\left(\frac{\partial^2\phi^-(x)}{\partial x^2}\right)\right]
$$

$$
\begin{aligned}
&\frac{NC_g}{2e^2}\frac{1}{\left(1 + \frac{C_g(N^2-1)}{12C_J}\right)}\left[\frac{\ddot{\phi}^+}{4} - \frac{e^2}{NC_g}\left(1 + \frac{C_g(N^2-6N+5)}{12C_J}\right)\left(\dot{p}^+(x+1) + \dot{p}^+(x-1)\right)\right] \\
&= E_J\left[1 - \frac{1}{8}a^2\left(\frac{\partial\phi^-(x)}{\partial x}\right)\right]\left[\frac{1}{4}a^2\left(\frac{\partial^2\phi^+(x)}{\partial x^2}\right)\right]
\end{aligned}
$$

- Taking just a single unit of the infinite current mirror qubit, we get the diagram like this.



Note that only the dotted joints mean that the wires intersect, else they just simply pass over. Rearranging this, this is nothing more than the 0-π qubit, as introduced by Kitaev, except for the inductors.

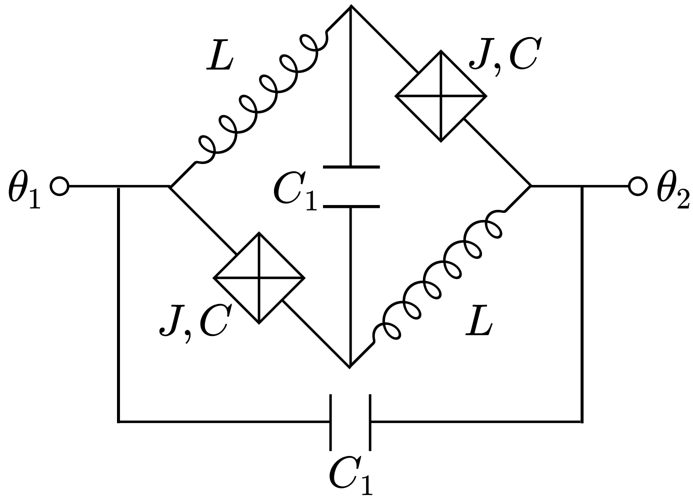


Image taken from [4]

This design is one of the highly analyzed ones due to its inherent protection and its easy to verify experimentally.

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